

Inter IIT Team Selection Test

Inter IIT Team Selection Test — ISRO Track

This is an individual assessment. Each candidate must work and submit independently. Do not discuss approaches, code, or write-ups with other candidates — submissions found to be substantially similar across candidates will be disqualified.

There are two tasks below. Both are compulsory.

Task 1: CLASS Lunar XRF Elemental Ratio Pipeline — Design & Visualization

Background

The Chandrayaan-2 Large Area Soft X-ray Spectrometer (CLASS) measures X-ray fluorescence (XRF) spectra excited by solar flares, from which elemental abundances and ratios (Mg/Si, Al/Si, Ca/Si, etc.) can be derived to map lunar surface composition. This task is based on the core methodology challenge behind Inter IIT Tech Meet 13.0's High Prep problem statement, "High-Resolution Elemental Mapping of Lunar Surface."

Provided Materials

- **Raw data — you will download this yourself.** Two CLASS footprints are specified below, one over a mare region and one over a highland region. For each, retrieve the corresponding **CLASS L1 spectrum** and the simultaneous **XSM solar spectrum** (same UTC window) from the CLASS data portal:

- Data portal: <https://pradan.issdc.gov.in/ch2/>

Footprint A: UTC 2022-04-15T05:36:40 to 2022-04-15T05:38:16, latitude 57.51°, longitude 78.14° **Footprint B:** UTC 2021-08-28T12:53:05 to 2021-08-28T12:54:41, latitude -30.52°, longitude 132.46°

You are responsible for locating, downloading, and correctly reading the right files for these windows — treat this as part of the task. No further guidance on the download process will be given.

- **Reference reading** (no further hints will be given beyond this):
 - CLASS instrument manual (available from the portal above)
 - Narendranath et al. (2024), "Lunar elemental abundances as derived from Chandrayaan-2," Icarus 410, 115898
 - Any other publicly available CLASS/XSM literature you find relevant

What You Need to Do

1. Visualize the raw data. Plot the raw spectra (counts vs. energy/channel) for both footprints, clearly labeled with axes, units, and which footprint each plot corresponds to. Add a brief written note (a few sentences) on anything you observe when comparing the two spectra.

2. Devise a detailed pipeline for deriving elemental ratios. This is a **methodology/design task, not a coding exercise**. We want a clear, step-by-step written plan for how you would go from the raw spectra to reliable elemental line ratios (Mg/Si, Al/Si, Ca/Si, and any others you judge

derivable). Pseudocode or short code snippets are welcome anywhere they help clarify a step, but the core deliverable is the written plan itself.

Your plan should explicitly address: - What must be done to the raw spectrum before line intensities can be trusted, and *why* each step is necessary. - Which elements/ratios can be reliably derived from CLASS data alone, which cannot, and your reasoning in each case. - How you would quantify and report uncertainty in your derived ratios. - Any external data or cross-checks you would use to validate your results.

Note: Superficial answers (e.g., “subtract the background, then fit Gaussians to the lines”) will not score well. We are evaluating the depth and correctness of your understanding of the instrument and technique — not general familiarity with spectral line-fitting.

Deliverables for Task 1

- Labeled plots of both spectra, with your written observations
 - Your pipeline design document
 - Any supporting code or pseudocode (optional)
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Task 2: GeoNLI — Satellite Image Captioning & Evaluation

Background

This task is based on Inter IIT Tech Meet 14.0’s Mid Prep problem statement, GeoNLI (“Natural Language Interpretation of Satellite Imagery”), which calls for a natural-language interface to satellite imagery supporting captioning, grounding, and visual question answering. This task focuses on the **captioning** component as a self-contained exercise.

Provided Materials

- **Raw data — you will locate this yourself.** Six specific images from the VRSBench dataset are specified below by ID. Using the dataset, retrieve the corresponding image file and its human-verified reference caption for each ID:

- Dataset: <https://huggingface.co/datasets/xiang709/VRSBench> (paper: arXiv:2406.12384)

Image IDs: P0003_0002.png , P0019_0002.png , P0060_0004.png , P0110_0017.png , P0146_0005.png , P0168_0009.png

You are responsible for figuring out which files in the dataset contain the images and which contain the reference captions, and for correctly matching them by ID. No further guidance on the dataset’s structure will be given.

- Reference benchmark for context: <https://vrsbench.github.io/>

What You Need to Do

1. Build a captioning pipeline using an **open, non-commercial-API** vision-language model — i.e., no calls to GPT-4V, Gemini, Claude, or similar hosted commercial APIs. This mirrors the actual constraint in the original problem statement.
2. Generate a caption for each provided image using your pipeline.
3. Compute the **BLEU score** of your generated captions against the provided reference captions, per image and in aggregate.
4. Write a short analysis (under one page) of your model’s failure modes, using specific examples from your own generated outputs — e.g., a missed object, an incorrect count, a wrong terrain classification, a hallucinated detail. Generic statements about VLM limitations without concrete examples will not score well.

Deliverables for Task 2

- Code for your captioning pipeline
- Generated captions and BLEU scores (per-image and aggregate)

- Your failure-mode analysis
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Submission Format & Guidelines

- **Submit individually** — one submission per candidate, covering both tasks.
- Package your submission as a **single zip file** named `<YourFullName>_ISRO_SelectionTest.zip`, containing:
 - `/task1/` — your plots, pipeline write-up (PDF or Markdown), and any supporting code
 - `/task2/` — your captioning code, generated captions with BLEU scores (a CSV or Markdown table is fine), and your failure-mode write-up
 - A top-level `README.md` briefly listing what's in each folder and how to run any code you've submitted
- Write-ups may be submitted as PDF, Markdown, or Word (.docx) — pick whichever you're comfortable with; content quality matters far more than formatting.
- Code should be runnable as-is or accompanied by clear setup instructions (e.g., a `requirements.txt`) — code that doesn't run and can't be easily fixed will be evaluated on design intent only, not on output correctness.